TARVI: Transcription-Factor Aided RNA Velocity Inference with Supervised Latent Time and Velocity-Pseudotime Blending

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***Abstract*—Reconstructing cell-fate trajectories from single-cell RNA sequencing (scRNA-seq) is fundamental to understanding cellular development and disease. RNA velocity infers cellular dynamics from unspliced-to-spliced mRNA ratios, but existing methods share three key limitations: transcription rates are static gene-level constants that ignore cell-state regulation; latent cell time is weakly supervised and poorly calibrated; and locally inferred velocity vectors are often inconsistent with global trajectories, producing reversed or discontinuous flows. We present TARVI (Transcription-Factor-Aided RNA Velocity Inference), a deep probabilistic framework that extends the VeloVI variational autoencoder to address all three limitations, through cell-specific transcription rates driven by transcription factor-target regulatory priors (ChEA/ENCODE), a supervised residual latent-time head, novel velocity-time consistency and temporal-smoothness losses, and a velocity-pseudotime blending step. Evaluated on 5 datasets against 4 baselines with 8 metrics, TARVI matches the strongest baseline on cross-boundary direction accuracy (CBDir: 0.933 vs. VeloVI's 0.931) while improving Spearman pseudotime correlation by 126% over VeloVI (0.747 vs. 0.331), recovering coherent directional flows and smooth developmental gradients where competing methods fail.**

***Index Terms*—RNA velocity, variational autoencoder, transcription factor, latent time, deep generative model**

I. Introduction

Reconstructing cell-fate trajectories from a static snapshot of transcriptomes is a central problem in single-cell biology [1], [2]: cells are destroyed during measurement, so the developmental state of every cell must be inferred from expression alone. *RNA velocity* [3], [4] reframes this as a local extrapolation problem by jointly counting *spliced* and *unspliced* mRNA per gene, yielding a vector field over the cell embedding whose streamlines approximate developmental trajectories without experimental time labels. Its biophysical basis is a system of linear ODEs per gene g [3], [4]:

$$\frac{du_g}{dt} = \alpha_g(t) - \beta_g\, u_g, \tag{1}$$

$$\frac{ds_g}{dt} = \beta_g\, u_g - \gamma_g\, s_g, \tag{2}$$

where α_g is the transcription rate, β_g the splicing rate, and γ_g the degradation rate, giving the velocity $v_g = ds_g/dt = \beta_g u_g - \gamma_g s_g$. At steady state ($du/dt = ds/dt = 0$) the on-state satisfies $u_g/s_g = \gamma_g/\beta_g$; deviations indicate active induction or repression.

Despite a decade of progress, contemporary estimators still suffer from three structural limitations: (i) the per-gene transcription rate α_g is a static scalar, ignoring cell-state-dependent regulation by transcription-factor (TF) networks; (ii) latent cell time, recovered as a by-product of mixture-state assignment, is weakly supervised and uncalibrated against developmental ordering; and (iii) locally inferred velocity vectors and globally observed cell trajectories are estimated independently, leading to reversed flows and instability under count splitting [5], [6].

We present TARVI (Transcription-Factor-Aided RNA Velocity Inference), a deep probabilistic framework that explicitly uses TF expression as a mechanistic input to modulate transcription rates, addressing all three gaps jointly. Our contributions are:

- A masked TF to gene weight matrix (ChEA/ENCODE priors) producing cell-specific $\boldsymbol{\alpha}_i$, jointly optimised end-to-end against the ODE likelihood.
- A residual latent-time head supervised by a stop-gradient ODE-time signal, with a graph-Laplacian temporal-smoothness regulariser, plus a velocity-time consistency loss that directly penalises the angle between each cell's velocity and its displacement toward later cells.
- A velocity-pseudotime blending step that fuses the supervised time head with velocity-graph diffusion pseudotime for a globally grounded temporal coordinate.
- Evaluation on five biologically diverse datasets against four baselines (Velocyto, scVelo dynamical, VeloVI, TFvelo) with 8 metrics, where TARVI matches the strongest baseline on cross-boundary direction accuracy while substantially improving pseudotime calibration.

II. Related Work

Velocyto [3] pioneered RNA velocity by fitting a steady-state slope γ_g/β_g via linear regression on the u–s phase portrait. While conceptually simple, it assumes a single global kinetic regime per gene and cannot capture the transient induction–repression dynamics that dominate differentiation. scVelo [4] resolved this limitation by solving the full ODE system (1)–(2) analytically for a two-state (induction/repression) model and inferring cell times via an Expectation Minimization (EM) algorithm. However, α_g remains a static gene-level constant that ignores cell-state-dependent transcriptional regulation, and the expensive per-gene EM fits scale poorly to large gene panels.

VeloVI [7], [8] replaced the EM procedure with a Variational Auto Encoder (VAE), enabling scalable amortised inference with uncertainty quantification across four kinetic states (induction, induction-steady, repression, repression-steady). Despite these advances, transcription rates in these models remain single learnable scalars with no regulatory input, and latent time, a by-product of mixture weights, is often poorly calibrated against developmental ordering. TFvelo [9] addressed the regulatory gap by modelling α_g as a static linear function of transcription factor expression, improving interpretability; however, TFvelo is not a generative model, provides no uncertainty quantification, and produces no latent cell embedding.

DeepVelo [10] and scTour [11] moved beyond closed-form ODE solutions by learning continuous transcriptional dynamics with neural ODEs [12], enabling flexible modelling of non-linear kinetics. More recently, VeloTrace [6] addressed the *velocity–trajectory gap* (the inconsistency between locally inferred velocity vectors and globally observed cell trajectories) by jointly optimising velocity and trajectory with NeuralODEs to ensure temporal coherence across the UMAP embedding.

Saelens et al. [1] established systematic trajectory inference benchmarking, and recent large-scale RNA velocity benchmarks [13], [14] have standardised community metrics including CBDir (cross-boundary direction accuracy) and ICCoH (in-cluster coherence of velocity vectors). Li et al. [5] further identify instabilities such as reversed velocity flows across current methods and introduce a replicate-coherence metric via count splitting, directly motivating the need for a supervised, calibrated time head. TARVI directly targets the limitations identified above: cell-specific TF-regulated α_g, a supervised residual time head for calibrated latent time, and a velocity–pseudotime blending step for global temporal grounding, with all three components integrated within a single probabilistic generative framework.

III. TARVI Methodology

A. Preprocessing

Standard preprocessing uses `scv.pp.moments` after QC, normalisation (10^4 counts/cell), log1p, and top-2000 HVG selection using the Scanpy/scVelo stack [4], [15]. TARVI additionally applies (a) per-gene MinMax scaling of `Ms`/`Mu` to $[0, 1]$; and (b) an r^2-based gene filter retaining only genes with positive scVelo r^2 and $\gamma > 0$.

B. Generative Model

Extending the VeloVI [7] framework, TARVI defines the following generative process for a cell i with scaled spliced and unspliced vectors $\mathbf{s}_i, \mathbf{u}_i \in \mathbb{R}^G$:

$$\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \tag{3}$$

$$(\boldsymbol{\rho}_i, \boldsymbol{\tau}_i, \boldsymbol{\pi}_i) = f_\theta^{\text{dec}}(\mathbf{z}_i), \tag{4}$$

$$\boldsymbol{\alpha}_i = \text{softplus}\big((W \odot M)\, \mathbf{s}_i^{\text{TF}} + \mathbf{b}\big), \tag{5}$$

$$(\mathbf{s}_i, \mathbf{u}_i) \sim \sum_{k=1}^{4} \pi_{i,g,k} \, \mathcal{N}\big(\mu_{i,g,k},\, \sigma_{g,k}^2\big). \tag{6}$$

The encoder q_ϕ maps $[\log(\mathbf{s} + 0.01) \| \log(\mathbf{u} + 0.01)]$ to $\mathbf{z}_i \in \mathbb{R}^d$ ($d = 10$); the decoder f_θ^{dec} produces induction/repression progress $\boldsymbol{\rho}_i, \boldsymbol{\tau}_i \in [0, 1]^G$ and mixture weights $\boldsymbol{\pi}_i \in \Delta^3$.

1) Four-State Kinetics and ODE Solutions: The four states are induction ($k = 1$), induction-steady ($k = 2$), repression ($k = 3$), and repression-steady ($k = 4$). The induction- and repression-branch Gaussian means $\mu_{i,g,k}^{(s,u)}$ follow VeloVI's closed-form solutions of (1)–(2) [7]: induction from $u{=}s{=}0$ over $t_{\text{ind}} = t_s \rho_{i,g}$, and repression from the induction endpoint over $t_{\text{rep}} = (t_{\max} - t_s)\tau_{i,g}$; the induction-steady and repression-steady means are $(\alpha_g/\beta_g,\, \alpha_g/\gamma_g)$ and $(0, 0)$. All rates are `softplus`-constrained and clamped to $[0, 50]$ with learnable t_s ($t_{\max} = 20$). With cell-specific $\boldsymbol{\alpha}_i$ the same closed forms hold, α_g broadcast to $[N \times G]$.

The ODE-derived velocity is computed per state as $v_g = \beta_g u - \gamma_g s$ and aggregated by the mixture weights (6). Fig. 1 shows the full architecture.

C. TF-Regulated Transcription Rate

Transcription rates are determined by hierarchical, modular gene regulatory networks (GRNs) [16]; TARVI reflects this biology by replacing the static gene-level α_g with a cell-specific rate computed from a masked, learnable weight matrix. A binary mask $M \in \{0, 1\}^{G \times K}$ is derived from ENCODE ChIP-seq peak annotations and ChEA 2016 TF–target associations [17]–[20], retaining the top $K \leq 99$ TFs per gene; weights are warm-started as $W_{gk}^{(0)} = 0.1 \cdot \rho_S(\mathbf{s}_g, \mathbf{s}_k)$. Since ChEA/ENCODE cover human and mouse, our five mouse/human benchmarks fall within their annotation scope, as also leveraged by TFvelo [9].

The TF-regulated transcription rate is:

$$\boldsymbol{\alpha}_i = \text{softplus}\big((W \odot M)\, \mathbf{s}_i^{\text{TF}} + \mathbf{b}\big), \quad \boldsymbol{\alpha}_i \in \mathbb{R}_{>0}^G. \tag{7}$$

The bias $\mathbf{b}$ is warm-started from VeloVI's basal α_{unconstr} (frozen), so the TF module adds an incremental cell-specific signal. L1 regularisation $\lambda_{\text{L1}} \|W\|_1 / (GK)$ promotes sparsity; unlike TFvelo, $\boldsymbol{\alpha}_i$ is jointly optimised end-to-end against the ODE likelihood.

D. Residual Latent-Time Head

VeloVI's cell time $t_i^{\text{ODE}} = \mathbb{E}_{k \sim \boldsymbol{\pi}_i}[t_k(\boldsymbol{\rho}_i, \boldsymbol{\tau}_i, t_s)]$ is discontinuous at state boundaries and weakly supervised, yielding poor developmental calibration (Section IV-B).

TARVI introduces a dedicated *residual time head*: a two-layer MLP on $\mathbf{z}_i$ computing $\mathbf{h}_1 = \text{ReLU}(W_1 \mathbf{z}_i) + W_{\text{skip}} \mathbf{z}_i$, $\mathbf{h}_2 = \text{ReLU}(\text{LN}(\mathbf{h}_1) W_2)$, and $t_i = \sigma(W_3 \mathbf{h}_2) \in [0, 1]$, where $\text{LN}(\cdot)$ denotes layer normalisation. The skip connection stabilises early training during the high-KL phase. The head is

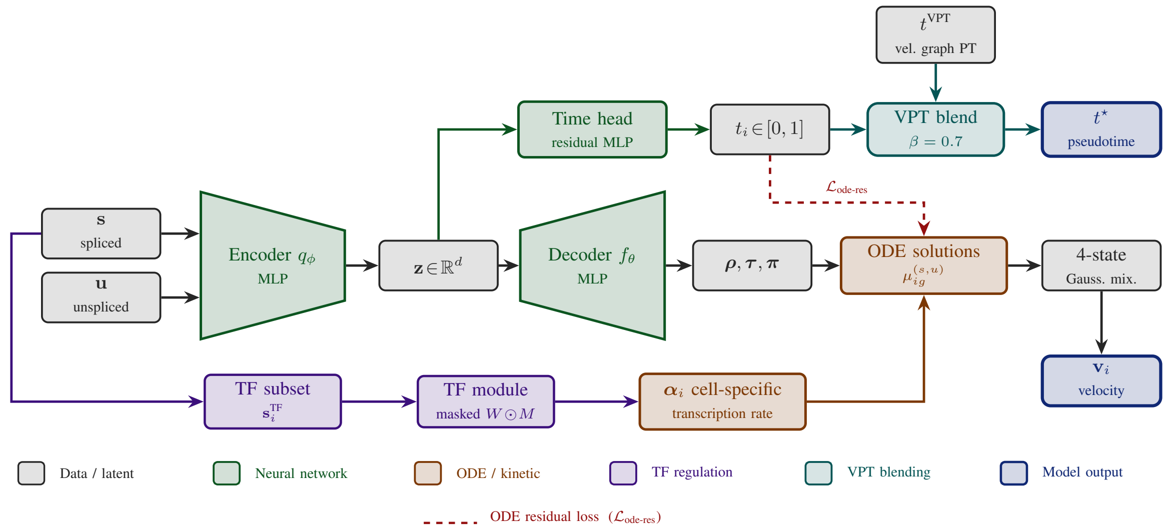


Fig. 1. TARVI architecture. Module colours follow the in-figure legend; the dashed red path indicates the ODE-residual loss $\mathcal{L}_{\text{ode-res}}$.

supervised by $\mathcal{L}_{\text{time}} = \text{MSE}(t_i,\ \text{sg}(t_i^{\text{ODE}}))$, where $\text{sg}(\cdot)$ denotes the stop-gradient operator, so t_i tracks t_i^{ODE} without distorting the kinetic parameters.

E. Training Losses

The total loss is:

$$\begin{aligned}\mathcal{L}_{\text{total}} &= \mathcal{L}_{\text{recon}} + w_{\text{kl}}\,\text{KL}(\mathbf{z}) + \text{KL}_{\text{cpu}}(\boldsymbol{\pi}) \\ &\quad + \lambda_{\text{time}}\mathcal{L}_{\text{time}} + \lambda_{\text{ode}}\mathcal{L}_{\text{ode-res}} \\ &\quad + \lambda_{\text{vtc}}\mathcal{L}_{\text{vtc}} + \lambda_{\text{ts}}\mathcal{L}_{\text{ts}} + \lambda_{\text{L1}}\mathcal{L}_{\text{L1}}. \end{aligned} \tag{8}$$

Reconstruction and KL: $\mathcal{L}_{\text{recon}}$ is the negative Gaussian mixture log-likelihood; $\text{KL}(\mathbf{z})$ is the standard VAE KL; $\text{KL}_{\text{cpu}}(\boldsymbol{\pi})$ is the VeloVI Dirichlet KL for mixture weights.

ODE-residual loss (scVelo-inspired): Given the learned time t_i, the ODE predictions $\hat{u}_{ig}, \hat{s}_{ig}$ are re-evaluated via a soft gate $g_{ig} = \sigma(5(t_{s,g} - t_i \cdot t_{\max}))$ that smoothly blends induction and repression branches:

$$\hat{u}_{ig} = g_{ig}\, u_{\text{ind}}(t_i) + (1 - g_{ig})\, u_{\text{rep}}(t_i), \tag{9}$$

$$\hat{s}_{ig} = g_{ig}\, s_{\text{ind}}(t_i) + (1 - g_{ig})\, s_{\text{rep}}(t_i), \tag{10}$$

$$\mathcal{L}_{\text{ode-res}} = \tfrac{1}{2}\left[\overline{\left(\tfrac{\hat{s}-s}{\bar{s}}\right)^2} + \overline{\left(\tfrac{\hat{u}-u}{\bar{u}}\right)^2}\right]. \tag{11}$$

Per-gene normalisation by $\bar{s}, \bar{u}$ prevents high-magnitude genes from dominating the loss. This is the differentiable analogue of scVelo's [4] EM time-update step. Default: $\lambda_{\text{ode}} = 1.0$.

Velocity–time consistency loss: A coherent RNA velocity field should point from earlier to later cells. For $n_p = \min(B/2, 128)$ random index pairs (i,j) per minibatch, sorted so $t_i \le t_j$:

$$\mathcal{L}_{\text{vtc}} = \frac{1}{n_p} \sum_{(i,j)} w_{ij} \cdot \text{ReLU}\big(-\cos(\mathbf{v}_i,\ \mathbf{s}_j - \mathbf{s}_i)\big), \tag{12}$$

where $w_{ij} = \text{clip}((t_j - t_i)/\overline{\Delta t}, 0, 3)$ upweights pairs with large temporal gaps. The hinge $\text{ReLU}(-\cos)$ penalises only wrong-pointing velocities, leaving correctly-aligned pairs unaffected. This is, to our knowledge, the first direct end-to-end pairwise penalty on the angle between velocity and the expression displacement vector. Default: $\lambda_{\text{vtc}} = 0.2$.

Temporal smoothness loss: To prevent the time head from over-fitting local noise, we impose a graph Laplacian regulariser over in-batch kNN pairs in latent space:

$$\mathcal{L}_{\text{ts}} = \frac{1}{Nk} \sum_i \sum_{j \in \mathcal{N}_i^z} (t_i - t_j)^2. \tag{13}$$

No precomputed graph is required; neighbourhood distances are computed on the fly. Default: $\lambda_{\text{ts}} = 0.1$.

Default weights: w_{kl}=1.0, η_{Dir}=0.25, λ_{time}=1.0, λ_{ode}=1.0, λ_{vtc}=0.2, λ_{ts}=0.1, λ_{L1}=0.01.

F. Velocity-Pseudotime Blending

The final coordinate $t_i^\star$ blends two complementary signals. After training, scVelo's diffusion pseudotime [4], [21] is run on the velocity transition graph to obtain $t_i^{\text{VPT}} \in [0,1]$, encoding global topology. The learned time t_i^{head} is then sign-aligned to t^{VPT} by Spearman correlation:

$$t_i^{\text{head}} \leftarrow \begin{cases} 1 - t_i^{\text{head}} & \rho_S(t^{\text{head}}, t^{\text{VPT}}) < 0, \\ t_i^{\text{head}} & \text{otherwise}, \end{cases}$$

and the two are convex-blended:

$$t_i^\star = \beta \cdot t_i^{\text{VPT}} + (1-\beta) \cdot t_i^{\text{head}}, \quad \beta = 0.7. \tag{14}$$

$\beta = 0.7$ was selected once by grid search over $\{0, 0.2, \ldots, 1.0\}$ on the pooled cross-dataset mean and held fixed for all five datasets (no per-dataset tuning); performance is stable across $\beta \in [0.5, 0.9]$. VPT alone inherits velocity errors near boundaries, while the time head alone may drift without topological grounding, which is why the blend consistently outperforms both individual signals.

TABLE I
BENCHMARKED DATASETS

Dataset	Biology	Source
Pancreas	Pancreatic endocrinogenesis	Bastidas-Ponce [22]
Bone marrow	Haematopoiesis	Setty et al. [23]
Forebrain	Neuronal lineages	La Manno et al. [3]
Chromaffin	Sympathetic nervous system	Furlan et al. [24]
scEU organoid	Metabolic labelling	dynamo [25]

TABLE II
EVALUATION METRICS (HIGHER IS BETTER FOR ALL)

Metric	Definition
ICCoH	Mean cosine sim. of each cell's unit velocity to its cluster-mean direction; measures within-cluster directional agreement.
CBDir	Cosine sim. between a cell's velocity and the mean velocity of its neighbours from the next developmental cluster; measures directional correctness at fate boundaries.
VelConf	Per-cell mean cosine sim. to kNN velocities; quantifies local field consistency.
PT-Spear	Spearman rank corr. of inferred pseudotime vs. geodesic UMAP distance from the root; captures global temporal ordering.
PT-DistCorr	Pearson corr. of pseudotime vs. pairwise UMAP distances; sensitive to linear distance structure.
PT-Cons	Fraction of kNN pairs whose pseudotime difference lies within 10% of Δt; measures local temporal smoothness.
RootAcc	Fraction of inferred root/terminal cells (from the velocity transition graph) matching biologically annotated progenitor or mature populations.
Gene-R^2_{spl}	Coefficient of determination between ODE-predicted and observed spliced counts; measures per-gene kinetic fit quality.

G. Training Procedure

TARVI is trained with AdamW ($\eta_0 = 10^{-2}$, weight decay 10^{-2}, gradient clip norm 10) for up to 500 epochs with early stopping on a 10% validation holdout (batch 256, or $\max(32, N/4)$ for small datasets). The VeloVI KL weight ramp is preserved; all auxiliary losses are active from epoch 0.

H. Datasets and Evaluation Protocol

TARVI and all four baselines are benchmarked across all five datasets (Table I); the pancreas dataset serves as the qualitative case study as it provides the most widely used, biologically annotated RNA-velocity trajectory [4], [7], making method differences immediately interpretable.

Baselines: Velocyto [3], scVelo dynamical [4], VeloVI [7], and TFvelo [9]. All methods are evaluated under identical preprocessing and evaluation conditions.

Metrics: Eight metrics [1], [13] are used. Gene-R^2_{spl} applies to TARVI and VeloVI only; see Table II.

IV. EXPERIMENTAL RESULTS

A. Quantitative Results

Table III reports full per-dataset results across all five benchmarks, and Table IV summarises cross-dataset means.

Key observations:

- **TARVI matches VeloVI on directional metrics and leads on temporal ones.** CBDir (0.933 vs. 0.931) and VelConf (0.927 vs. 0.920) are on par, while PT-Spear (0.747, $+126\%$), PT-DistCorr (0.771 vs. 0.341), and PT-Cons (0.986) are clearly top, confirming VPT blending yields a well-calibrated temporal coordinate.
- **TFvelo's ICCoH advantage comes at a large directional cost.** Despite the highest ICCoH (0.933), TFvelo's CBDir collapses to 0.569, indicating that static TF regression without a generative framework produces internally consistent but directionally incorrect velocity fields.
- **scVelo dynamical's** ICCoH, CBDir, and VelConf are undefined (shown as "–") because its per-gene EM algorithm produces NaN velocity values for genes whose kinetic parameters fail to converge, rendering all three cosine-based metrics undefined. Pseudotime metrics (PT-Spear, PT-DistCorr, PT-Cons) remain valid because scVelo's transition-probability graph filters NaN genes before computing pseudotime.
- **RootAcc is TARVI's weakest axis, driven by bone marrow.** Mean RootAcc (0.671) trails VeloVI (0.723) and scVelo (0.814) almost entirely because of bone marrow (0.493 vs. $\sim$0.93), where the local field is still strong (CBDir 0.947, VelConf 0.937). The miss lies in the post-hoc root/terminal extraction from the velocity graph, which is brittle on haematopoiesis's multifurcating topology, making macrostate-based terminal detection a clear next step.

Fig. 2 visualises the cross-dataset means. TARVI achieves the largest area on all temporal metrics (PT-Spear, PT-DistCorr, PT-Cons), tying VeloVI on velocity-quality (CBDir, VelConf). TFvelo's ICCoH spike collapses at CBDir, confirming directional inconsistency; scVelo dynamical's three cosine metrics are zeroed (NaN in table) while its RootAcc remains strong. A full component ablation (TF, VTC, and VPT blending across all five datasets) is provided as supplementary material, available at https://github.com/chandula00/TARVI/blob/main/supplementary/supplementary_material.pdf.

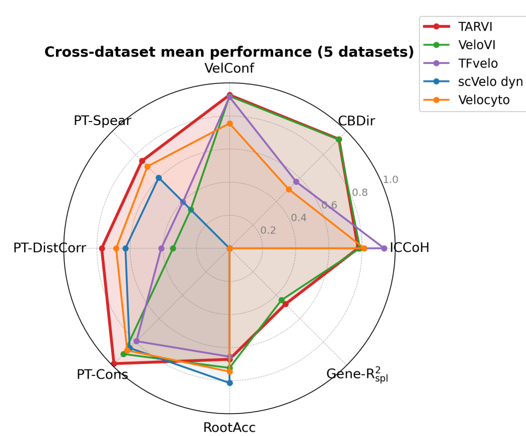


Fig. 2. Cross-dataset mean performance radar (5 datasets). Each axis is one metric on $[0, 1]$; larger area is better.

TABLE III

PER-DATASET FULL COMPARISON. BEST VALUE IN EACH DATASET/METRIC BLOCK IS BOLD; "–" MARKS UNDEFINED OR NON-APPLICABLE ENTRIES. SEE SECTION III-H FOR METRIC DEFINITIONS.

Dataset	Method	ICCoH	CBDir	Vel Confidence	PT Spearman	PT DistCorr	PT Consistency	Root Cell Acc	Gene-R^2
bonemarrow	TARVI	0.770	0.947	**0.937**	**0.418**	**0.412**	**1.000**	0.493	**0.350**
	VeloVI	0.804	**0.948**	0.936	0.094	0.116	0.979	**0.936**	0.235
	TFvelo	**0.950**	0.026	0.917	0.360	0.324	0.874	0.907	–
	scVelo_dyn	–	–	–	0.348	0.318	0.704	0.934	–
	velocyto	0.800	0.078	0.679	0.306	0.244	0.654	0.642	–
chromaffin	TARVI	0.780	**0.911**	0.924	0.807	0.826	**0.955**	0.917	**0.817**
	VeloVI	0.774	**0.911**	**0.925**	0.147	0.139	0.827	0.889	0.808
	TFvelo	**0.922**	0.005	0.883	0.052	0.027	0.640	0.778	–
	scVelo_dyn	–	–	–	0.706	0.696	0.809	**1.000**	–
	velocyto	0.816	0.057	0.758	**0.853**	**0.857**	0.808	0.972	–
forebrain	TARVI	0.825	0.920	0.911	**0.887**	**0.882**	**0.975**	0.610	**0.636**
	VeloVI	0.848	0.938	0.927	0.342	0.339	0.772	0.442	0.597
	TFvelo	**0.938**	**0.946**	**0.941**	0.424	0.384	0.957	**0.983**	–
	scVelo_dyn	–	–	–	0.697	0.685	**0.975**	0.802	–
	velocyto	0.823	0.852	0.857	0.883	0.859	0.974	0.779	–
pancreas	TARVI	0.721	0.944	0.939	**0.860**	**0.871**	**1.000**	**1.000**	0.498
	VeloVI	0.750	0.951	0.941	0.757	0.738	0.997	**1.000**	**0.501**
	TFvelo	**0.930**	**0.971**	**0.967**	0.783	0.783	0.933	0.531	–
	scVelo_dyn	–	–	–	0.799	0.793	0.981	**1.000**	–
	velocyto	0.815	0.837	0.865	0.834	0.851	0.988	**1.000**	–
sceu_organoid	TARVI	0.783	**0.945**	**0.926**	**0.765**	**0.862**	**0.998**	0.334	**0.084**
	VeloVI	0.761	0.908	0.870	0.316	0.375	0.944	**0.347**	0.074
	TFvelo	**0.927**	0.899	0.867	0.368	0.543	0.568	0.076	–
	scVelo_dyn	–	–	–	0.470	0.647	0.789	0.334	–
	velocyto	0.813	0.702	0.618	0.623	0.609	0.943	0.334	–

TABLE IV

CROSS-DATASET MEAN PERFORMANCE OVER THE FIVE BENCHMARKS (BOLD: BEST AMONG ALL COMPARED METHODS; "–": UNDEFINED)

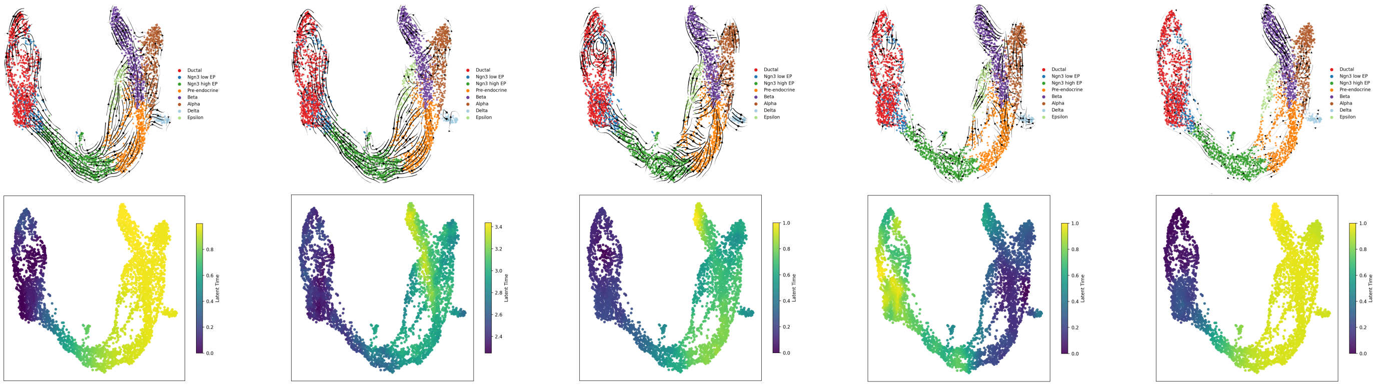
Method	ICCoH↑	CBDir↑	VelConf↑	PT-Spear↑	PT-DistCorr↑	PT-Cons↑	RootAcc↑	Gene-R^2_{spl}↑
TARVI	0.776	**0.933**	**0.927**	**0.747**	**0.771**	**0.986**	0.671	**0.477**
VeloVI	0.787	0.931	0.920	0.331	0.341	0.904	0.723	0.443
TFvelo	**0.933**	0.569	0.915	0.397	0.412	0.794	0.655	–
scVelo dynamical	–	–	–	0.604	0.628	0.852	**0.814**	–
Velocyto	0.813	0.505	0.755	0.700	0.684	0.873	0.745	–

B. Streamline and Latent-Time Analysis

We compare all five methods on the pancreatic endocrinogenesis benchmark (Ductal $\rightarrow$ Ngn3 low/high $\rightarrow$ Alpha/Beta/Delta/Epsilon). *Streamlines (Fig. 3, top):* TARVI produces clean directional flow from Ductal through Ngn3 intermediates to all four mature populations; VeloVI recovers the same topology but with spurious streamlines near Ductal; scVelo dynamical shows circular artefacts at the kinetic switch; TFvelo and Velocyto yield only sparse, discontinuous fields. *Latent time (Fig. 3, bottom):* TARVI gives a smooth $0 \rightarrow 1$ gradient (Ductal ≈ 0, mature ≈ 1); VeloVI is uncalibrated ($\sim[2.4, 3.4]$); scVelo dynamical is smooth but lower-contrast; TFvelo's gradient is reversed (mature earlier than progenitor); Velocyto is correctly oriented but with weaker cluster separation. Quantitatively (Table III), TARVI achieves the best PT-Spear (0.860), PT-DistCorr (0.871), PT-Cons (1.000), and RootAcc (1.000); TFvelo leads on cosine metrics but its RootAcc collapses to 0.531, mirroring the reversed time gradient, the local-vs-global tension that TARVI's TF-regulated $\boldsymbol{\alpha}_i$ and VPT-blended time head are designed to resolve.

V. CONCLUSION

TARVI is a principled, modular deep generative model for RNA velocity that integrates transcription-factor regulation, a supervised residual time head, ODE-aligned auxiliary losses, and velocity-pseudotime blending into a single variational framework. While it builds on the VeloVI generative backbone, TARVI's architecture, training objectives, and biological grounding are fundamentally different: every component that drives its performance is either new or a substantive redesign of an existing idea. Across five biologically diverse datasets and four baseline methods, TARVI matches the strongest baseline on cross-boundary direction accuracy and leads on pseudotime quality, with a 126% improvement in Spearman pseudotime correlation over VeloVI. The complete implementation, benchmark harness, and evaluation suite are available online (https://github.com/chandula00/TARVI.git).



(a) TARVI (b) VeloVI (c) scVelo dynamical (d) TFvelo (e) Velocyto

Fig. 3. Pancreas UMAP, one method per column. Top row: velocity streamlines, with cells coloured by annotated cell type and arrows depicting the projected RNA velocity field. Bottom row: inferred latent time, from early to late.

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